

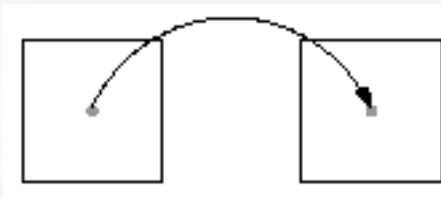


Konvolucija

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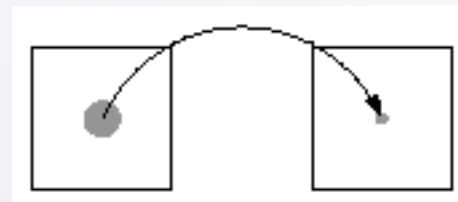
Operacije na digitalnoj slici

- Operacije zasnovane na točki
 - Izlazna vrijednost na osnovu vrijednosti piksela ulazne slike na istoj koordinati



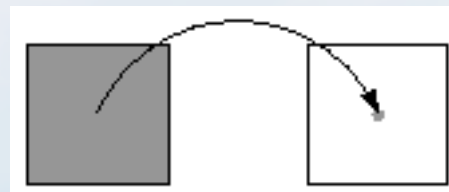
- **Lokalne operacije**

- Izlazna vrijednost temeljem ulazne na istoj koordinati i vrijednosti susjednih elemenata



- **Globalne operacije**

- Izlazna vrijednost ovisi o svim pikselima ulazne slike



Kontinuirani i diskretni signali

- Jednodimenzionalni kontinuirani signal:
 $f(x)$
- Diskretni jednodimenzionalni signal:
 $u(n)$
- Kontinuirani dvodimenzionalni signal:
 $f(x, y)$
- Diskretni dvodimenzionalni signal - digitalna slika:
 $u(m, n)$ (ili $u(m, n, i)$)

 m, n – koordinate u matrici (koordinate piksela)

Dvodimenzionalni linearni sustavi

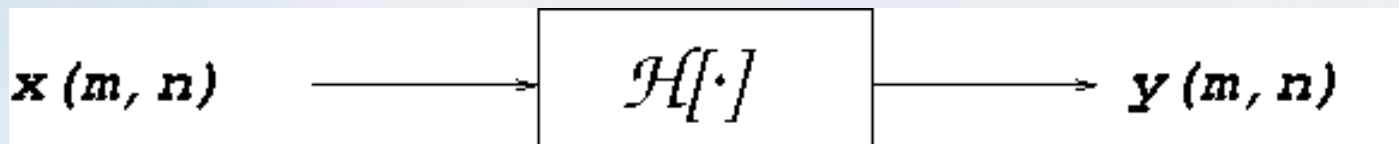
$x(m, n)$ - ulazna sekvenca

$y(m, n)$ - izlazna sekvenca:

$$y(m, n) = \mathcal{H}(x(m, n))$$

Sistem je linearan ako i samo ako za proizvoljne konstante a_1 i a_2 vrijedi linearna superpozicija:

$$\begin{aligned}\mathcal{H}(a_1 x_1(m, n) + a_2 x_2(m, n)) &= a_1 \mathcal{H}(x_1(m, n)) + a_2 \mathcal{H}(x_2(m, n)) \\ &= a_1 y_1(m, n) + a_2 y_2(m, n)\end{aligned}$$



Impulsni odziv sustava

- Odziv na dvodimenzionalnu Kronecker delta funkciju na lokaciji (m', n')

$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

$$\delta(m, n) = \delta(m)\delta(n)$$

$$h(m, n; m', n') = \mathcal{H}[\delta(m - m', n - n')]$$

U obradi slike

- Ulazna slika je idealna točka na koordinatama (m', n')

Vrijedi:

$$\delta(m, n) = \begin{cases} 1 & m = 0, n = 0 \\ 0 & m \neq 0 \text{ ili } n \neq 0 \end{cases}$$

$$x(m, n) = \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} x(m', n') \delta(m - m', n - n')$$

Izlaz bilo kojeg sustava moguće je dobiti iz ulaza i impulsnog odziva sustava:

$$x(m, n) = \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} x(m', n') \delta(m - m', n - n')$$

$$y(m, n) = \mathcal{H}[x(m, n)]$$

$$= \mathcal{H} \left[\sum_{m'} \sum_{n'} x(m', n') \delta(m - m', n - n') \right]$$

$$= \sum_{m'} \sum_{n'} x(m', n') \mathcal{H} [\delta(m - m', n - n')]$$

$$\Rightarrow y(m, n) = \sum_{m'} \sum_{n'} x(m', n') h(m, n; m', n')$$

Prostorno invariantni dvodimenzionalni linearni sustavi

- Translacija ulazne sekvence rezultira translacijom izlazne sekvence
 - **Za impuls u ishodištu vrijedi:**

$$\mathcal{H}[\delta(m, n)] = h(m, n; 0, 0)$$

$$\begin{aligned} h(m, n; m', n') &= \mathcal{H} [\delta(m - m', n - n')] \\ &= h(m - m', n - n'; 0, 0) \end{aligned}$$

$$\Rightarrow h(m, n; m', n') = h(m - m', n - n')$$

Impulsni odziv prostorno invariantnih linearnih sustava je funkcija varijabli pomaka

Konvolucija

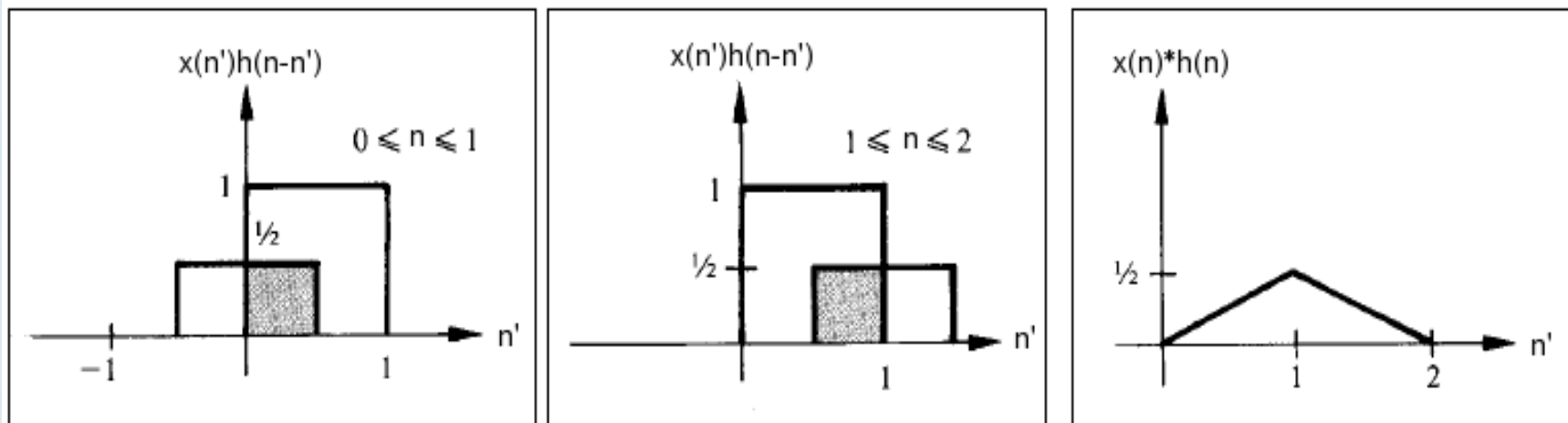
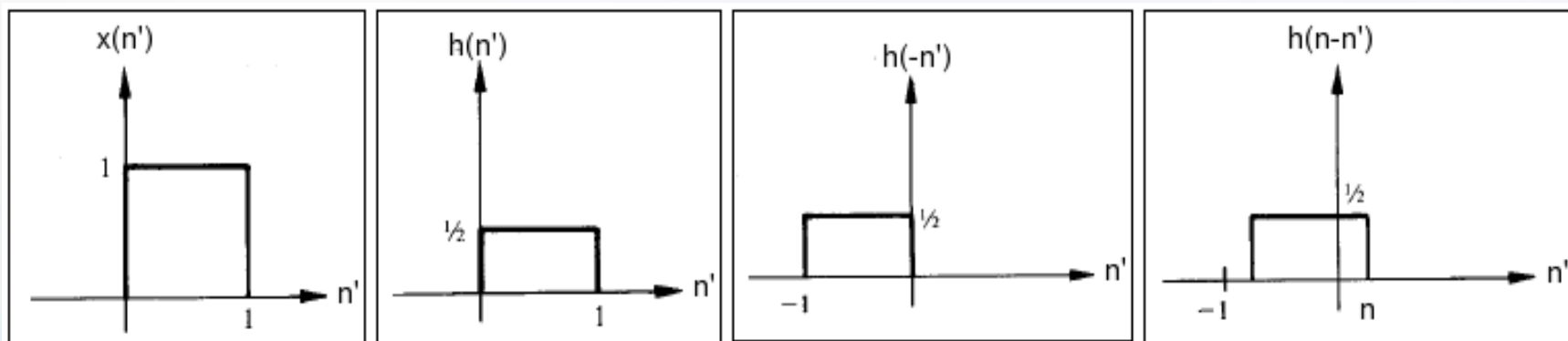
- Za prostorno invarijantne linearne sustave izlazna sekvenca $y(m, n)$ postaje:

$$y(m, n) = \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} h(m - m', n - n') x(m', n')$$

Gornji izraz nazivamo konvolucija ulaza sa impulsnim odzivom

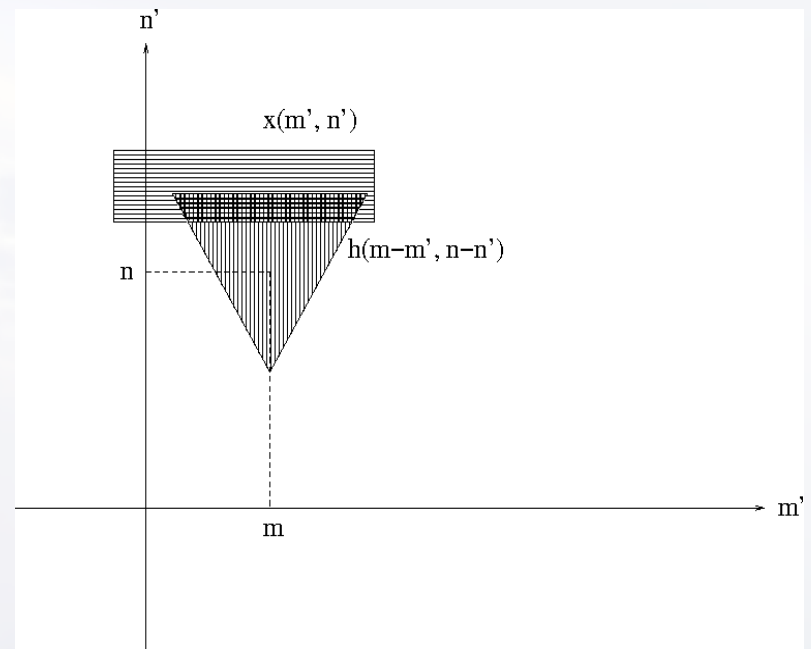
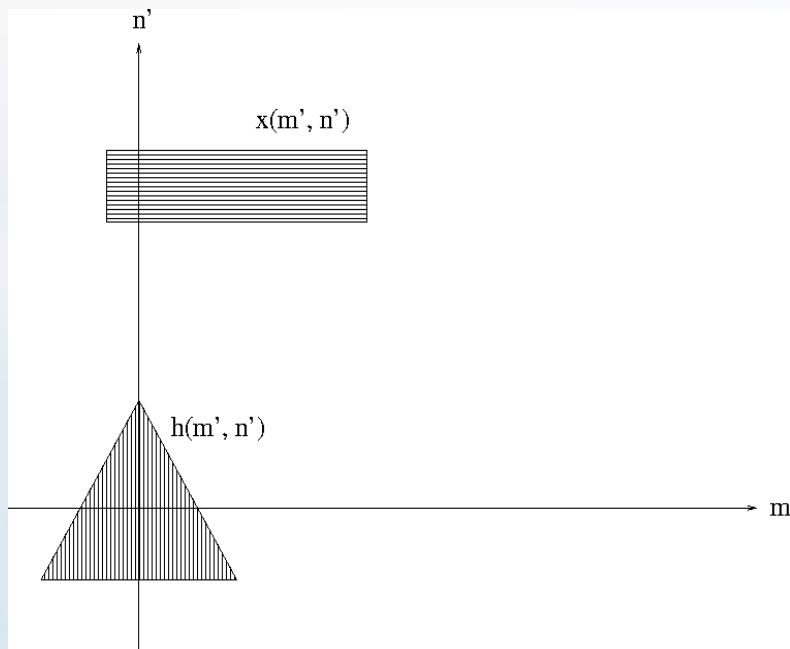
Konvolucija jednodimenzijskih signala

$$y(n) = \mathcal{H}[x(n)] = x(n) * h(n) = \sum_{n'} h(n - n')x(n')$$



Konvolucija dvodimenzionalnih signala

$$y(m, n) = \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} h(m - m', n - n') x(m', n')$$



- Izlazna vrijednost $y(m, n)$ u točki (m, n) predstavlja sumu produkata u području preklapanja

Konvolucija dvodimenzionalnih signala

$$y(m, n) = \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} h(m - m', n - n') x(m', n')$$

- Rezultat konvolucije (izlazna sekvenca $y(m, n)$) u točki (m, n) predstavlja sumu produkata $\{x(:, :)\}$ i $\{h(:, :)\}$ zarotiranog za 180° u području preklapanja

Linearna i cirkularna konvolucija (1D slučaj)

LINEARNA KONVOLUCIJA:

Neka je $x(n)$ 1D sekvenca širine (*region of support*) N i $h(n)$ sekvenca širine M

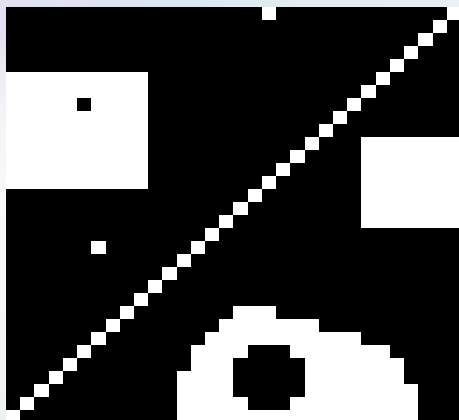
- izlazna sekvenca $y(n)$ širine $M+N-1$

CIRKULARNA KONVOLUCIJA:

Neka je $x(n)$ 1D periodička sekvence perioda N te $h(n)$ sekvenca širine $M < N$

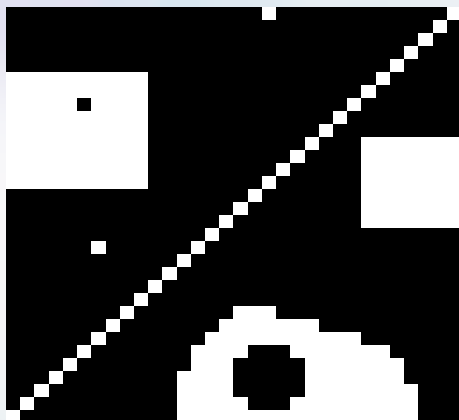
- Izlazna sekvenca $y(n)$ je periodička perioda N , a dobija se tako da se $h(n)$ popuni nulama do širine N te se koristi kao periodička istog perioda

2D konvolucija - primjer 1 (Kernel s pomakom)

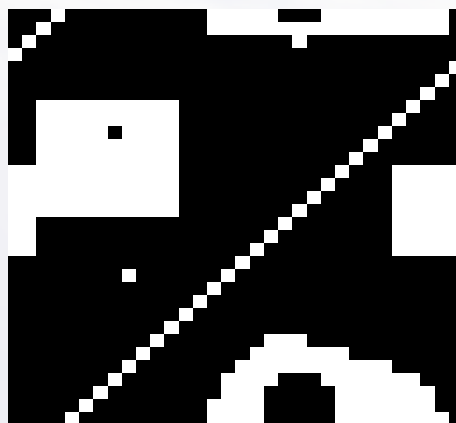
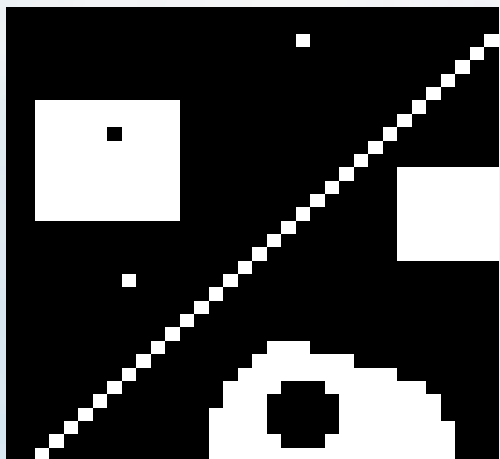


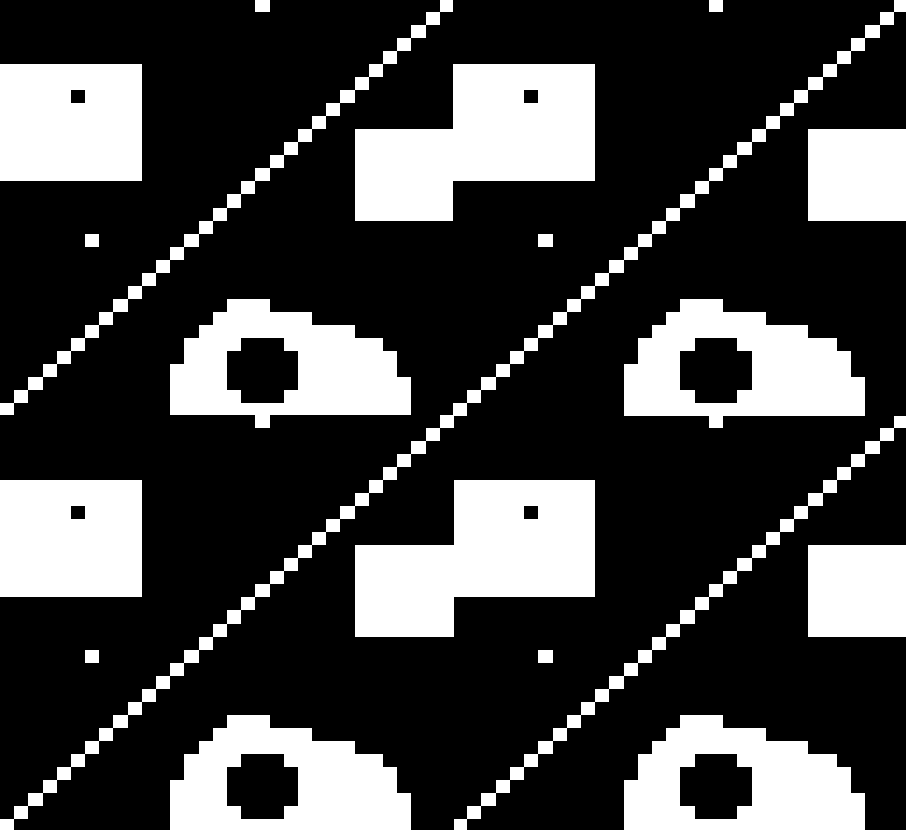
0	0	0
0	0	0
0	0	1

2D konvolucija - primjer 1 (Kernel s pomakom)

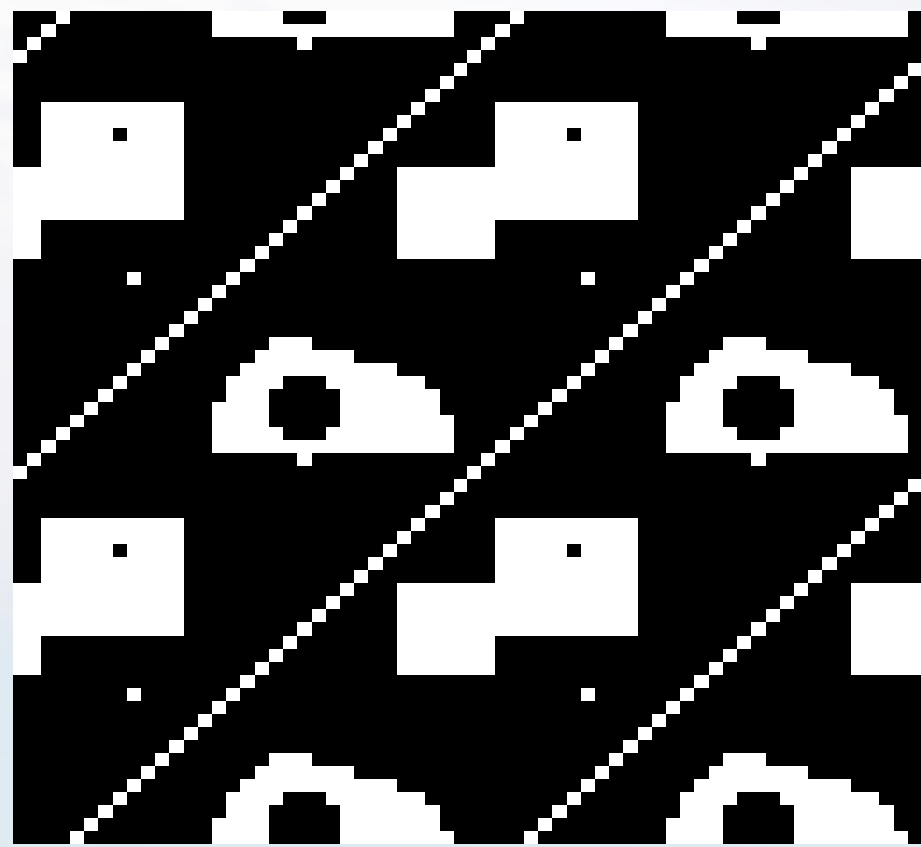


0	0	0
0	0	0
0	0	1

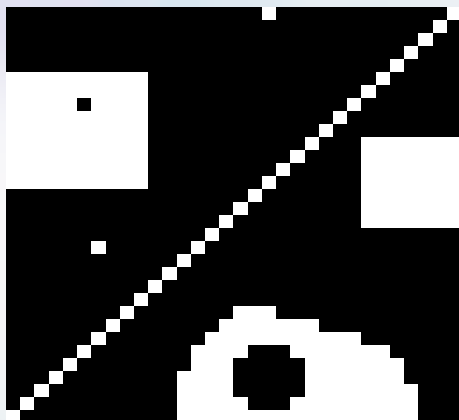




0	0	0
0	0	0
0	0	1

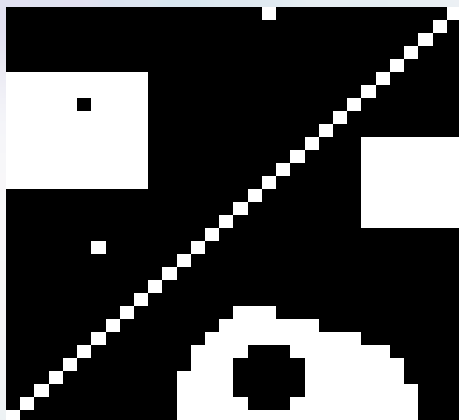


2D konvolucija - primjer 2 (Ujednačavanje)

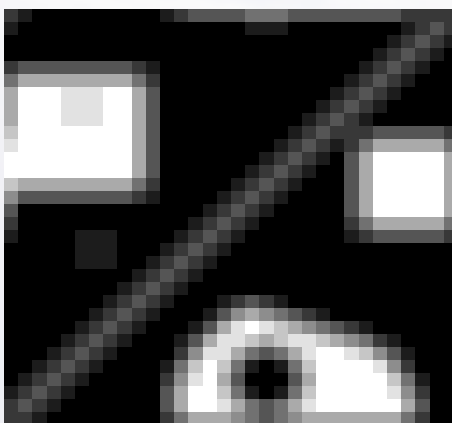
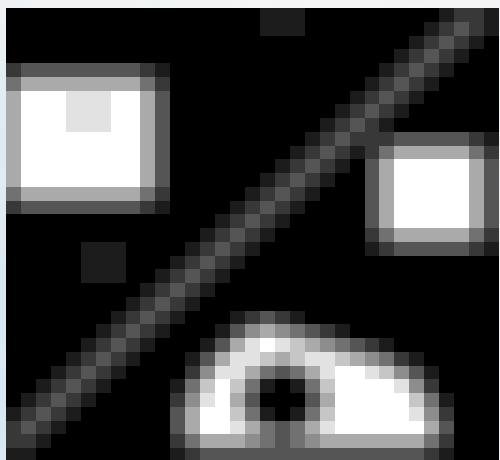


0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

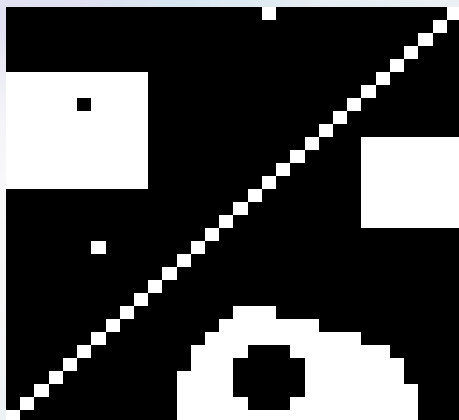
2D konvolucija - primjer 2 (Ujednačavanje)



0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

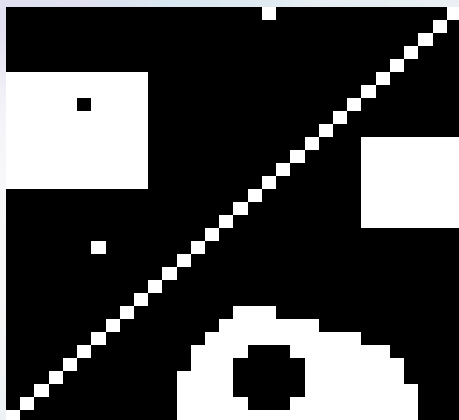


2D konvolucija - primjer 3 (Usmjereni Sobel operator)

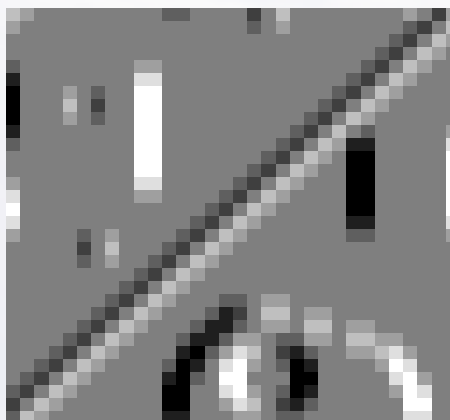
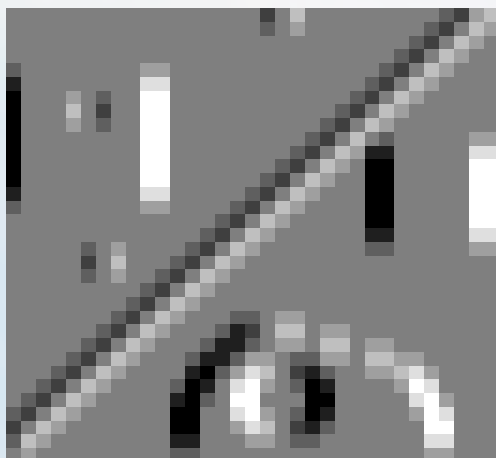


-1	0	1
-2	0	2
-1	0	1

2D konvolucija - primjer 3 (Usmjereni Sobel operator)



-1	0	1
-2	0	2
-1	0	1



Dvodimenzionalna konvolucija

- Jedna od najvažnijih operacija digitalne obrade slike
 - Temelj za mnogobrojne metode obrade slike

Osnovna ideja:

- Prozor konačnih dimenzija putuje preko izvorne slike
- Izlazne vrijednosti piksela predstavljaju težinsku sumu ulaznih piksela, gdje su težine zadane matricom prozora

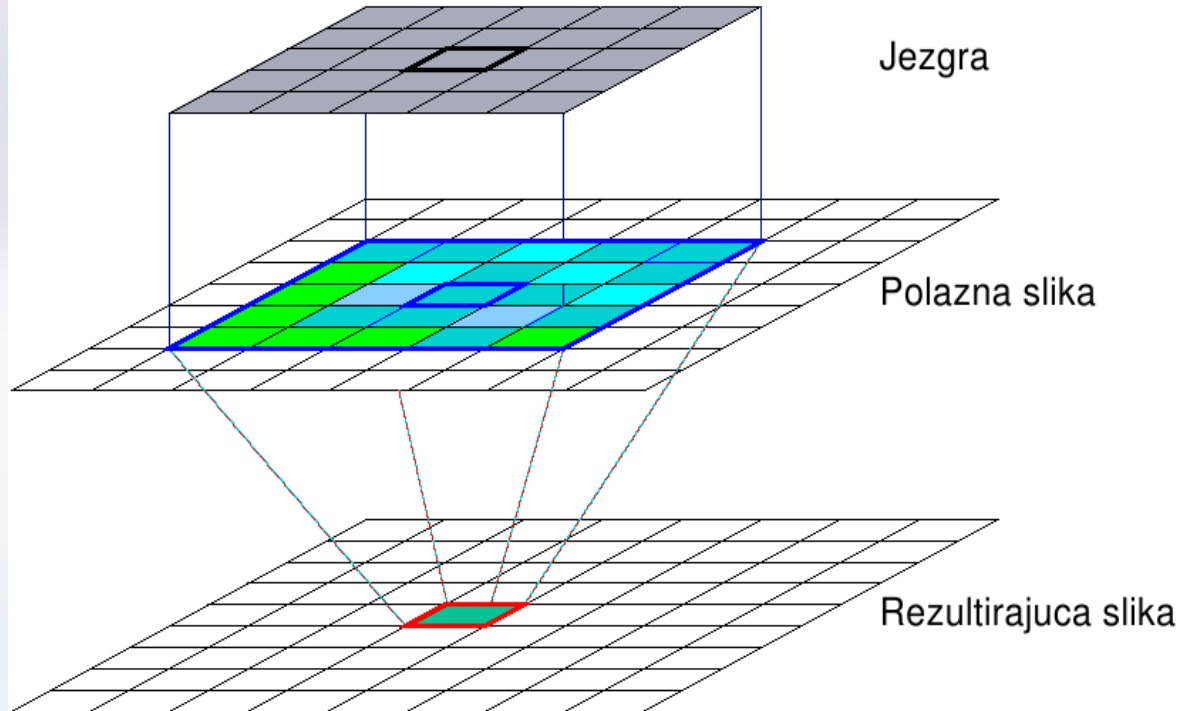
$$c[m, n] = a[m, n] * h[m, n] = \sum_{j=0}^{J-1} \sum_{k=0}^{K-1} h[j, k] a[m-j, n-k]$$

Filter $h[m, n]$

- Diskretna (kvadratna) jezgra neparne dimenzije
- Za filter $h[j, k]$ dimenzija $J*K$ koordinata $[j=0, k=0]$ smještena je u centru matrice h

$$h = \frac{1}{25} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{matrix} k=0 \\ j=0 \end{matrix}$$

- Broj operacija za dimenziju prozora $K*K$ je $O(K^2)$



- **Lokalna operacija**
- **Vrijednosti prozora (filtra) jednake nuli izvan pravokutnog prozora**

$$c[m, n] = a[m, n] * h[m, n] = \sum_{j=0}^{J-1} \sum_{k=0}^{K-1} h[j, k] a[m - j, n - k]$$

$$y(m, n) = \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} h(m - m', n - n') x(m', n')$$

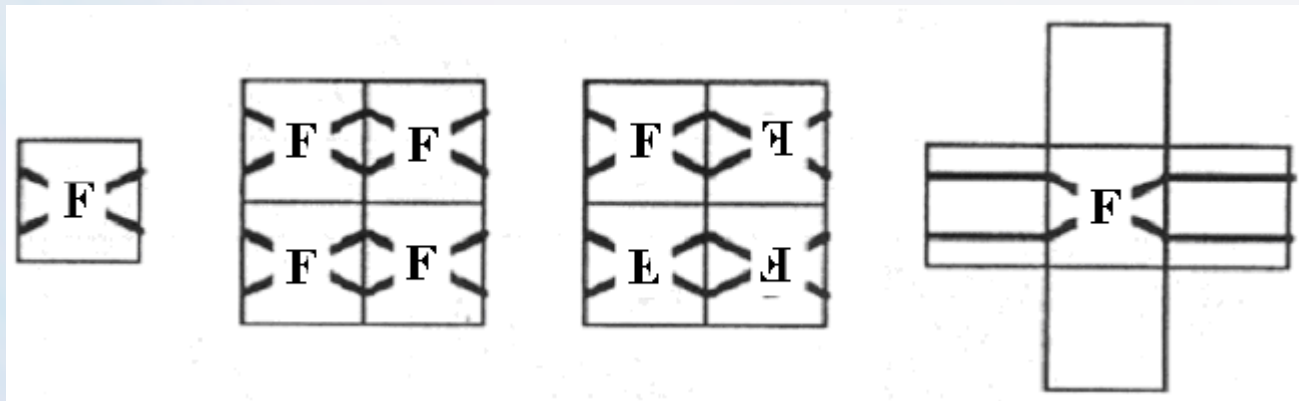
Konvolucija u rubnim djelovima slike

$$c[m, n] = a[m, n] * h[m, n] = \sum_{j=0}^{J-1} \sum_{k=0}^{K-1} h[j, k] a[m-j, n-k]$$

Izračunavanje konvolucije u točki $[m=0, n=0]$ zahtjeva vrijednosti $a[j, k]$ koje su van granica slike a

Nadomještanje vrijednosti:

- **Popunjavanje zadanom vrijednosti, periodičko dopunjavanje, zrcalno dopunjavanje, korištenje granične vrijednosti piksela**



Filtriranje slike

- Filter je operator dimenzija $N \times M$ zadan sa:
 - koeficijentima $w(i, j)$
 - koordinatama ishodišta $w(0, 0)$
- Odziv filtra računa se konvolucijom:
 1. Rotiranje filtra za 180°
 2. Postavljanje ishodišta iznad prvog piksela slike, računanje sume umnožaka koeficijenata i piksela slike

Korak 2 ponavlja se za sve piksele slike

3x3 filter:

$w(-1, +1)$	$w(0, +1)$	$w(+1, +1)$
$w(-1, 0)$	$w(0, 0)$	$w(+1, 0)$
$w(-1, -1)$	$w(0, -1)$	$w(+1, -1)$

Niskopropusno filtriranje:

- Ističu se niske frekvencije (umanjivanje naglih promjena vrijednosti)
 - vrijednost piksela približava se vrijednostima susjednih piksela (smoothing)
 - Izlazna slika može biti zamućena

Srednja vrijednost:

$1/9 *$

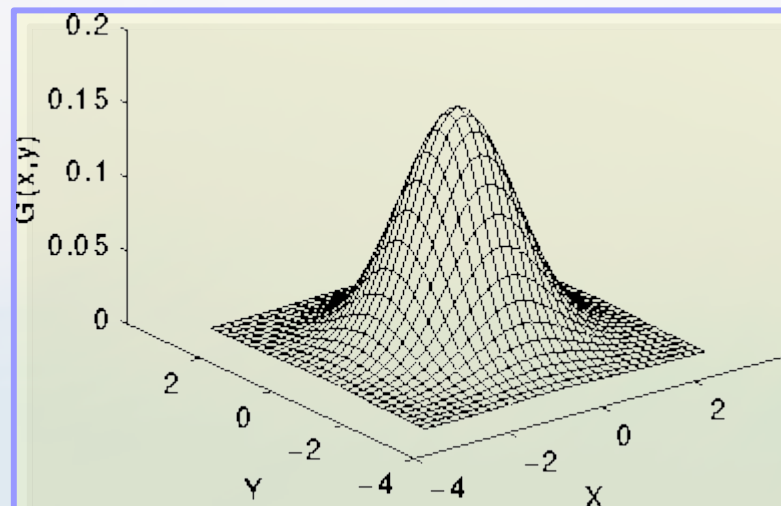
1	1	1
1	1	1
1	1	1

Gaussov filter:

$1/16 *$

1	2	1
2	4	2
1	2	1

2-D Continuous Gaussian filter



Niskopropusno filtriranje - primjer

Ulazna slika (dodano 10% šuma)



Usrednjavanje filtrom dimenzija 5x5



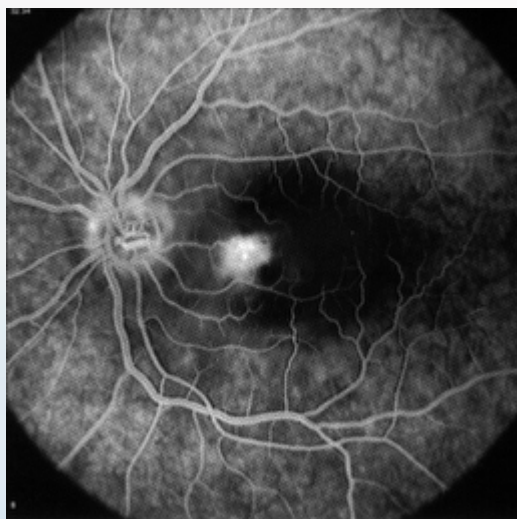
Visokopropusno filtriranje

- Ističe visoke frekvencije (nagle skokove među vrijednostima susjednih piksela)

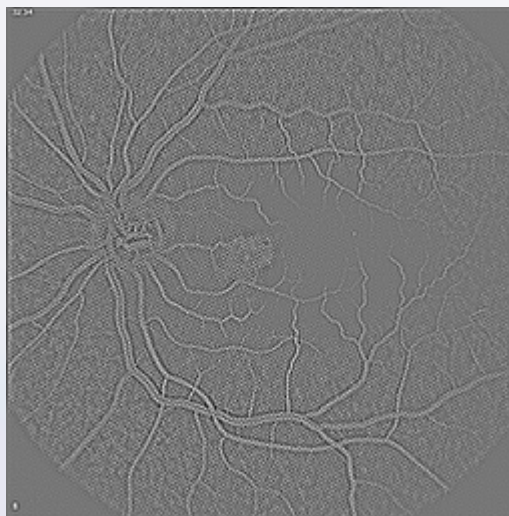
Image Sharpening:

Laplace visokopropusni filter:

0	-1	0
-1	4	-1
0	-1	0



filtrirana slika



original+filtrirana slika



Detekcija rubova

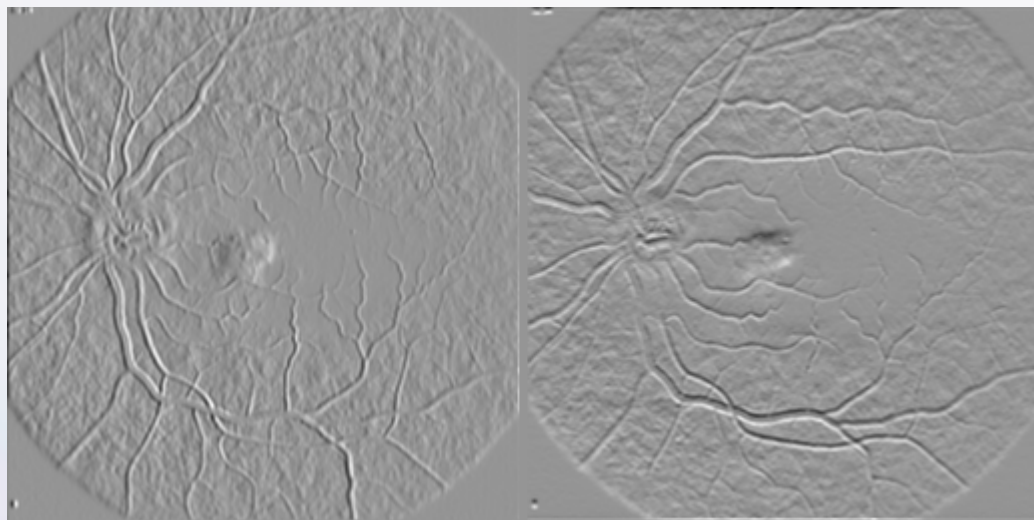
- **Gradijent: vektor (usmjerena vrijednost)**
- **Mjeri iznos promjene (prvu derivaciju) u ortogonalnim smjerovima**



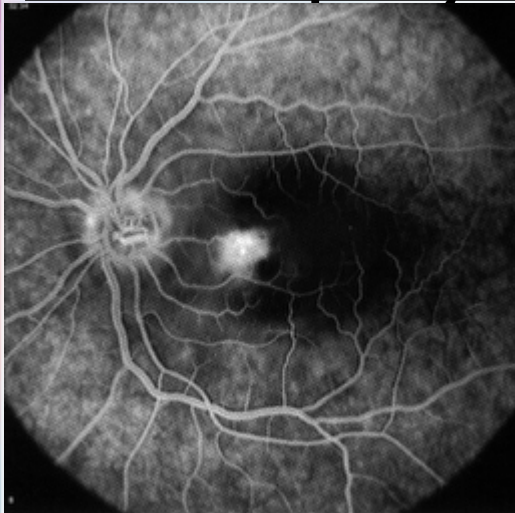
Sobel Operators

$$\begin{array}{|c|c|c|} \hline -1 & 0 & 1 \\ \hline -2 & \boxed{0} & 2 \\ \hline -1 & 0 & 1 \\ \hline \end{array} \quad \begin{array}{|c|c|c|} \hline -1 & -2 & -1 \\ \hline 0 & \boxed{0} & 0 \\ \hline 1 & 2 & 1 \\ \hline \end{array} \quad \begin{array}{c} \downarrow \\ \downarrow \\ \downarrow \end{array}$$

→ direction of gradients



Nelinearna operacija: magnituda promjene



Sobel Operators

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & \boxed{0} & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & \boxed{0} & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

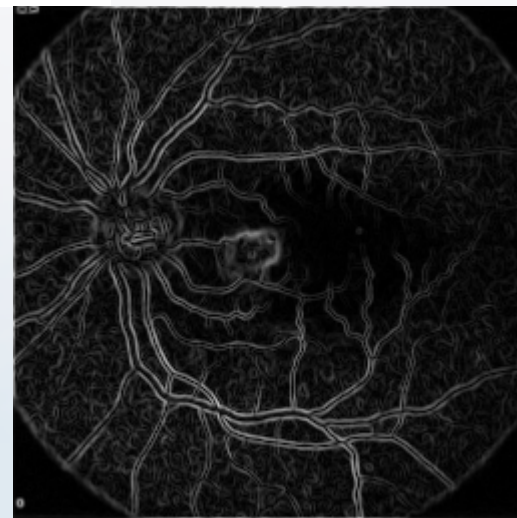
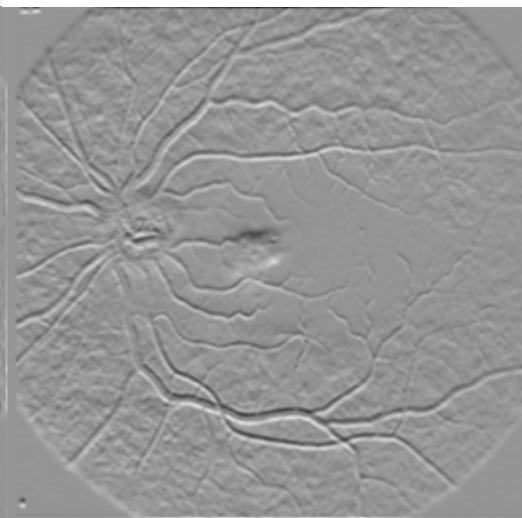
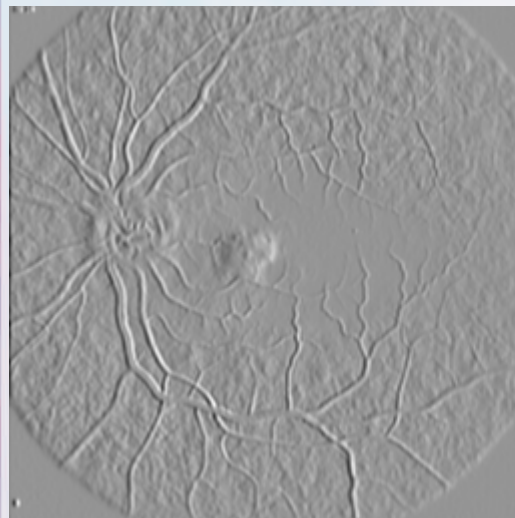


direction of gradients

$$G = G_x \vec{x} + G_y \vec{y}$$

$$|G| = \sqrt{G_x^2 + G_y^2}$$

$$\theta = \arctan \frac{G_x}{G_y}$$

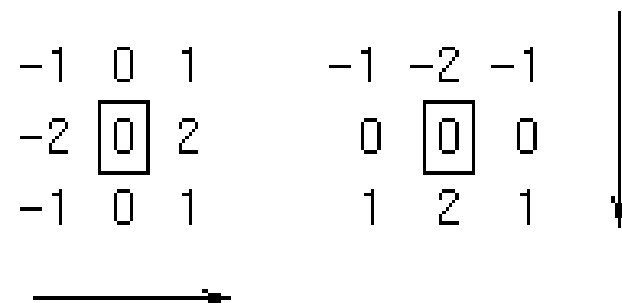


Gradijent operatori

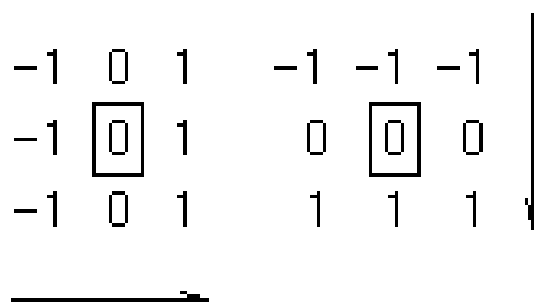
Roberts Operators

$$\begin{array}{cc} \boxed{0} & 1 \\ -1 & 0 \end{array} \quad \begin{array}{cc} \boxed{1} & 0 \\ 0 & -1 \end{array}$$

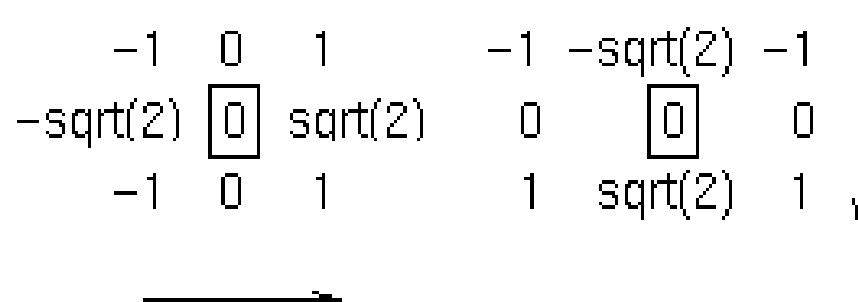

Sobel Operators

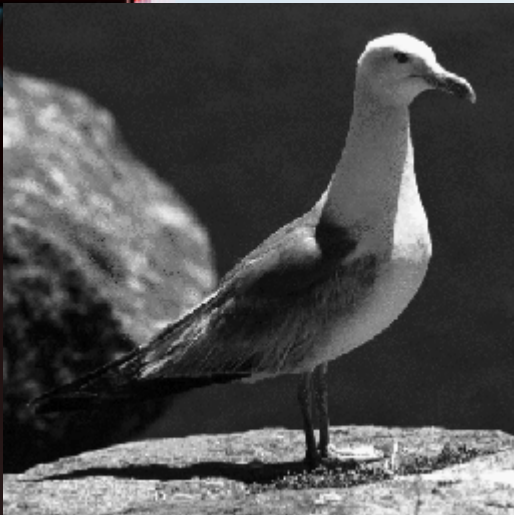
$$\begin{array}{ccc} -1 & 0 & 1 \\ -2 & \boxed{0} & 2 \\ -1 & 0 & 1 \end{array} \quad \begin{array}{ccc} -1 & -2 & -1 \\ 0 & \boxed{0} & 0 \\ 1 & 2 & 1 \end{array}$$


Prewitt Operators

$$\begin{array}{ccc} -1 & 0 & 1 \\ -1 & \boxed{0} & 1 \\ -1 & 0 & 1 \end{array} \quad \begin{array}{ccc} -1 & -1 & -1 \\ 0 & \boxed{0} & 0 \\ 1 & 1 & 1 \end{array}$$


Isotropic Operators

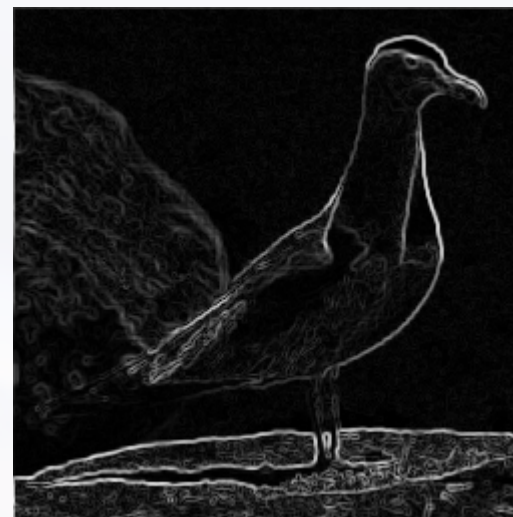
$$\begin{array}{ccc} -1 & 0 & 1 \\ -\sqrt{2} & \boxed{0} & \sqrt{2} \\ -1 & 0 & 1 \end{array} \quad \begin{array}{ccc} -1 & -\sqrt{2} & -1 \\ 0 & \boxed{0} & 0 \\ 1 & \sqrt{2} & 1 \end{array}$$




Roberts



Sobel



Prewitt



Isotropic

